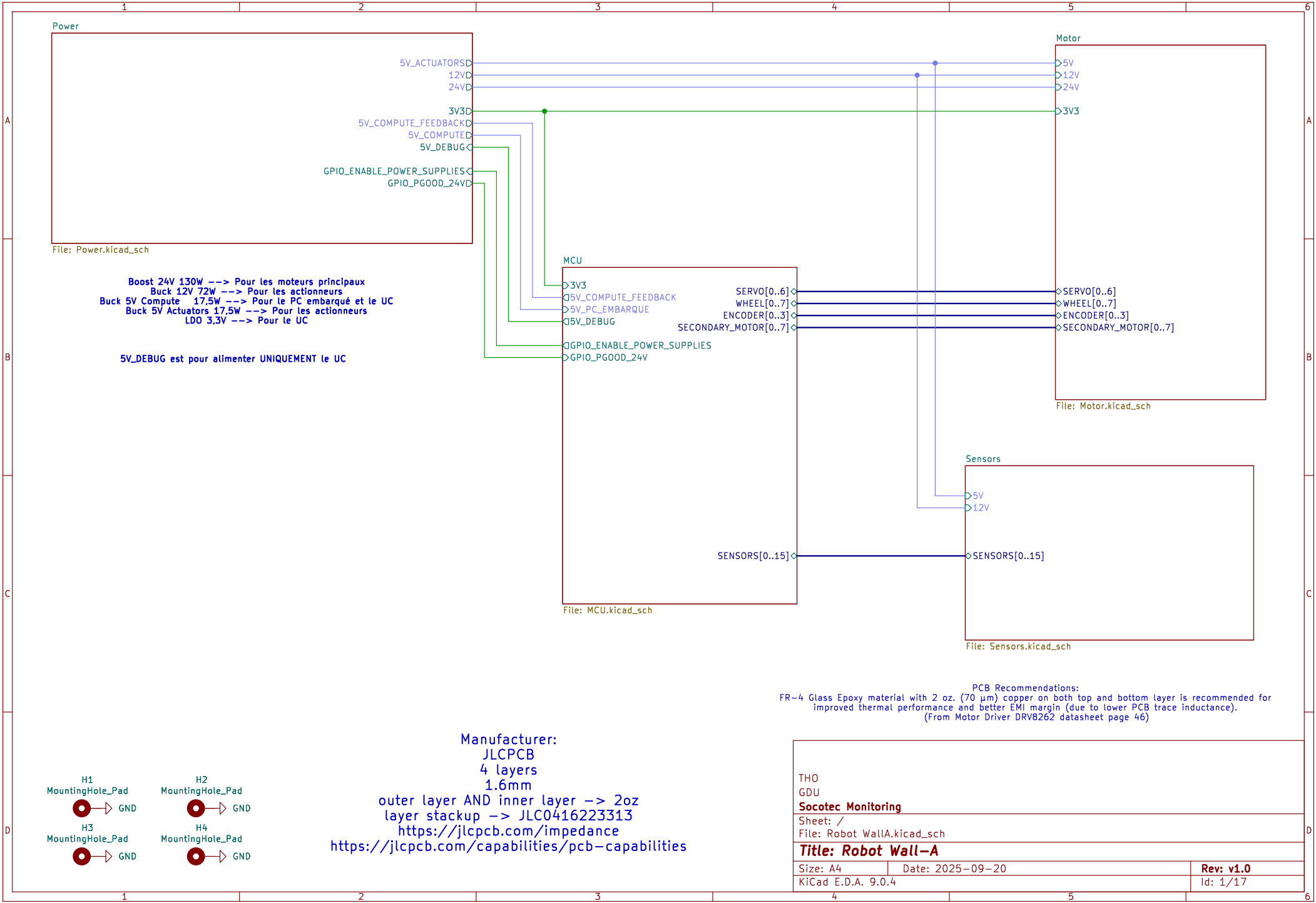
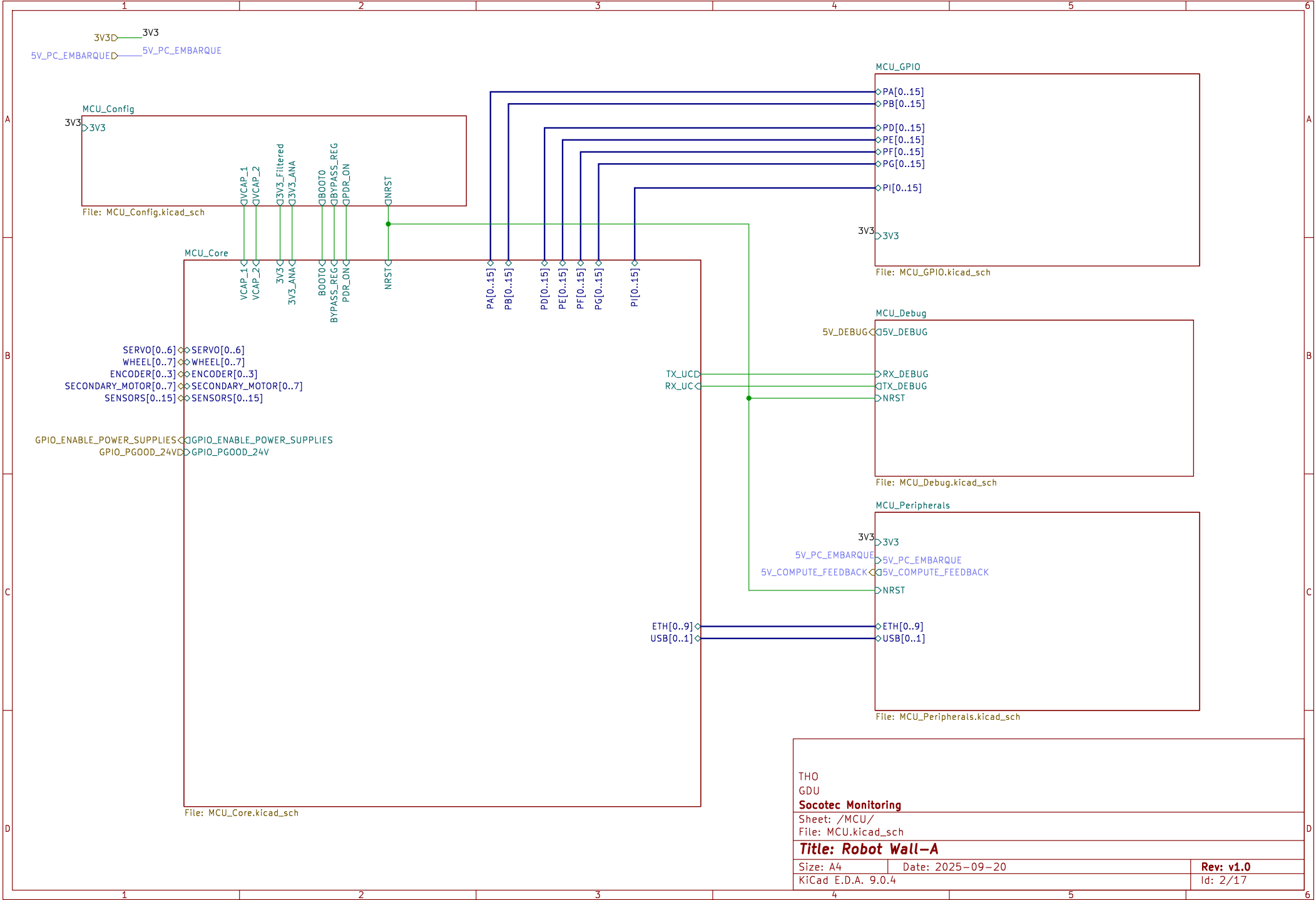






TH0
GDU

Socotec Monitoring

Sheet: /MCU/
File: MCU.kicad_sch

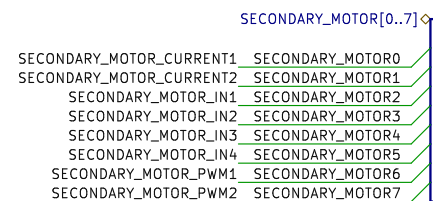
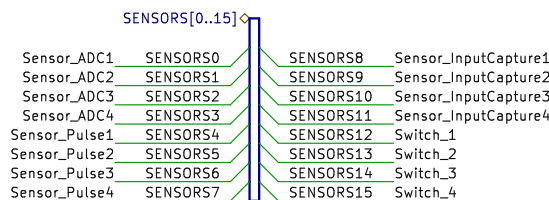
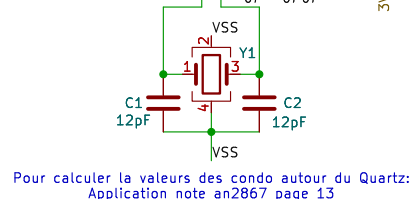
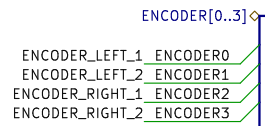
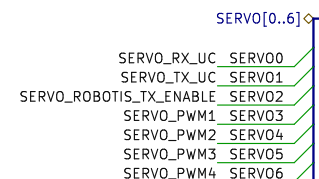
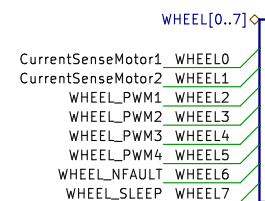
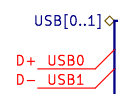
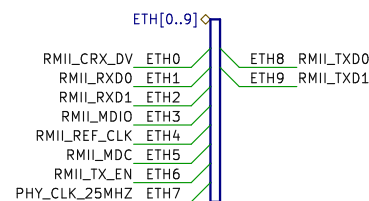
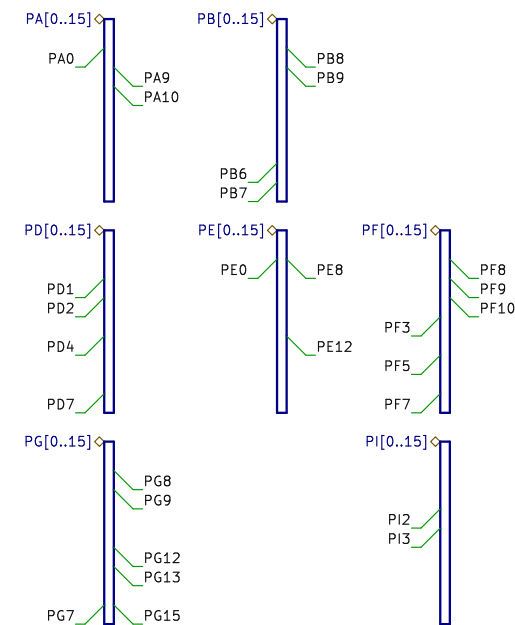
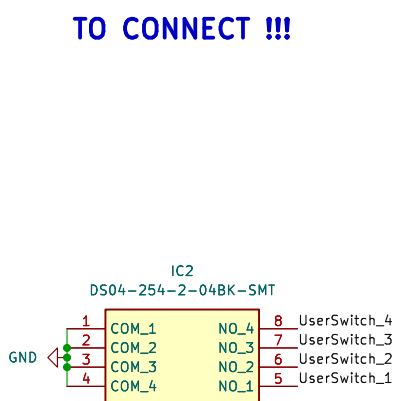
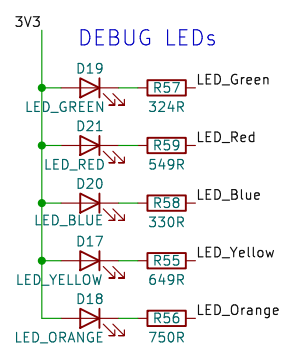
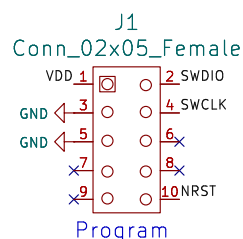
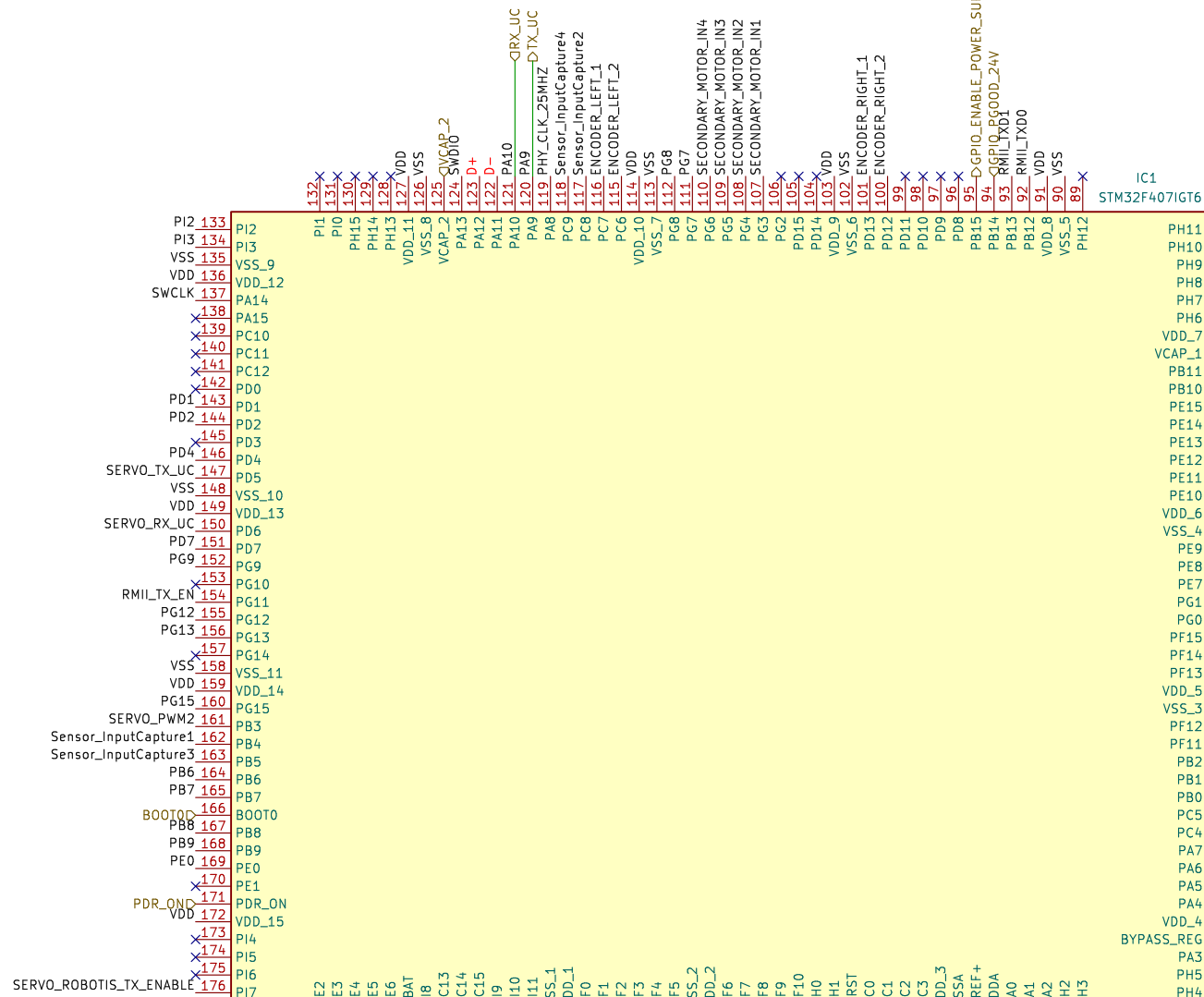
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

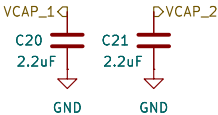
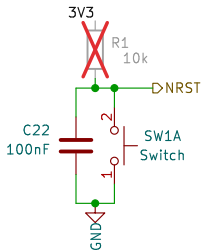
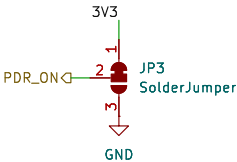
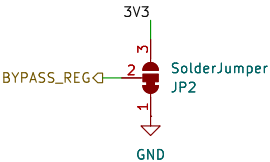
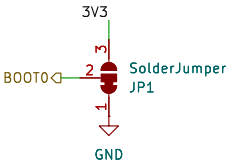
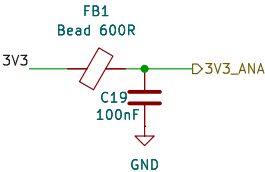
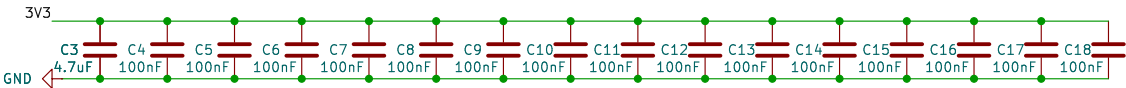
Rev: v1.0

Id: 2/17



Must be as close as possible from μC

3V3 $\rightarrow$ 3V3_Filtered



TH0
GDU

Socotec Monitoring
Sheet: /MCU/MCU_Config/
File: MCU_Config.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20
KiCad E.D.A. 9.0.4

Rev: v1.0
Id: 4/17

Reproduction de la doc "MB1137" d'un µC STM32 (pdf dans le git, dossier Docs)

USB

5V_PC_EMBARQUED 5V
3V3 3V3
NRST NRST
ETH[0..9]

FB2 FerriteBead
3V3 3V3_Filtered

5V_USB HighCurrent_3A
USB C - PC Embarqué
J2 USB4216-03-A

5V_COMPUTE_FEEDBACK 5V Diode 3A
D22
5V_USB
D+ USB0
D- USB1
IC3 USBLC6-4SC6
D+ 1 I/O1 I/O4 6 CC1
2 GND VBUS 5 5V_USB
3 I/O2 I/O3 4 CC2

CC1 CC2
R2 5.1k R3 5.1k

5V_USB A1 GND_1 CC2 B5 CC2
A4 VBUS_1 DP2 B6 D+
CC1 A5 CC1 B7 D-
D+ A6 DP1 SBU2 B8
D- A7 DN1 GND_3 G1
A8 SBU1 GND_4 G2
B1 GND_2 GND_5 G3
B4 VBUS_2 GND_6 G4

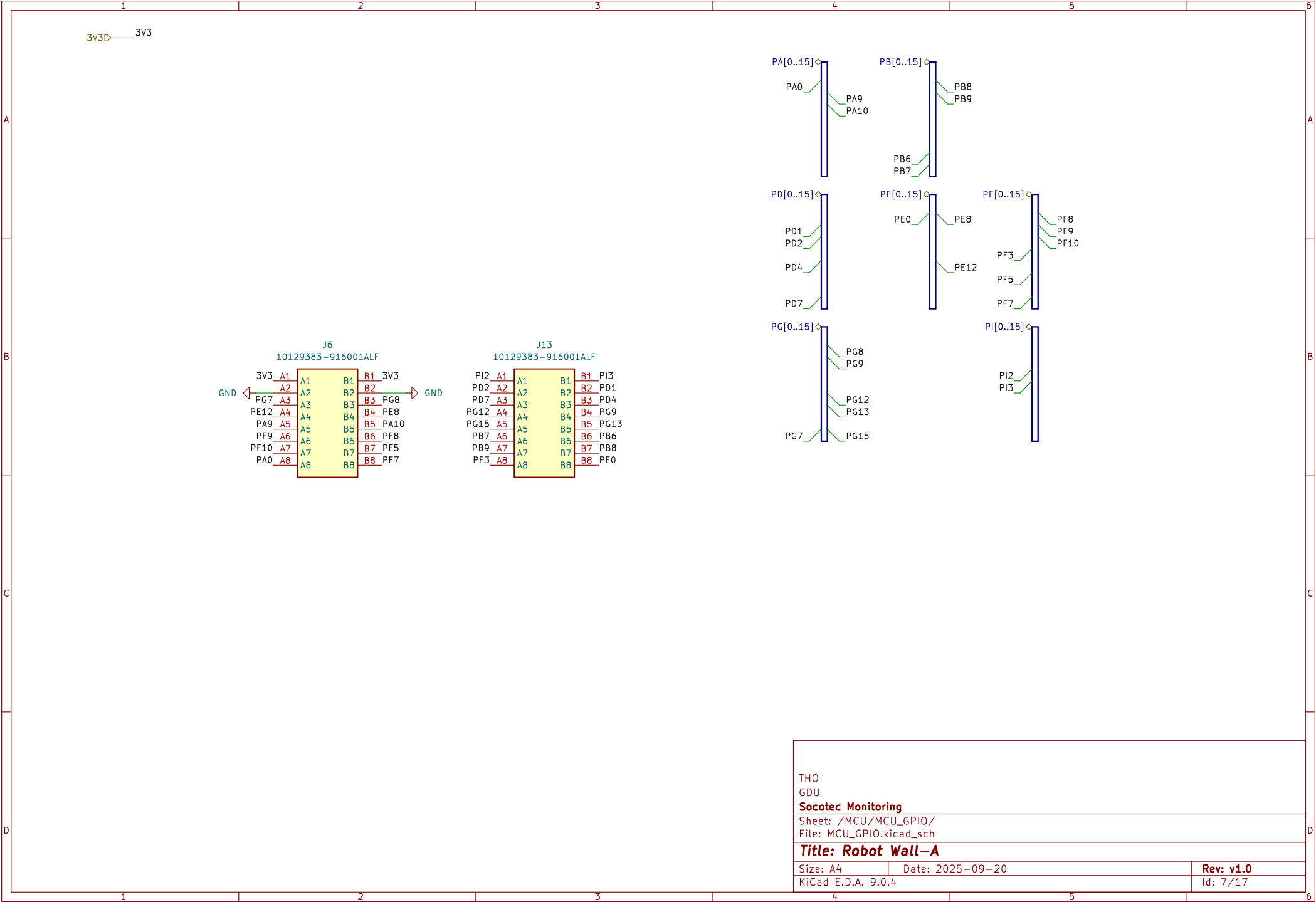
TX peut être connecté sur RX car le module est compatible auto-MDIX
T1 H1102NL
3V3_Filtered 1 RXP 2 TD+ TX+ 16 TX+
3 RXN 3 TD- TX- 15
4 NC_1 NC_5 13
5 NC_2 NC_4 12
6 RD+ RX+ 11 RX+
7 CT_2 NC_3 10 RX-
8 RD- RX- 9 RX-
3V3_Filtered 1 C29 100nF 2 GND
3V3_Filtered 1 C30 100nF 2 GND
R22 75R R23 75R R24 75R R25 75R
JP5 Open JP11 Open
GND

J3 SS-90000-009
TX+ 1 8
TX- 2 9
RX+ 3 10 LED_Green
4 11
5 12 LED_Yellow
6 MH3
7 MH4
R19 270R R20 10k R21 270R
GND GND GND

D+ DiffPair_90R
D- DiffPair_90R
TX+ DiffPair_100R
TX- DiffPair_100R
RX+ DiffPair_100R
RX- DiffPair_100R
TXP DiffPair_100R
TXN DiffPair_100R
RXP DiffPair_100R
RXN DiffPair_100R
Tuto tune differential pairs
<https://www.youtube.com/watch?v=M13QxtPvXY&t=798s>

TH0
GDU
Socotec Monitoring
Sheet: /MCU/MCU_Peripherals/
File: MCU_Peripherals.kicad_sch
Title: Robot Wall-A
Size: A4 Date: 2025-09-20 Rev: v1.0
KiCad E.D.A. 9.0.4 Id: 5/17

Id: 6/17



TH0

GDU

Socotec Monitoring

Sheet: /MCU/MCU_GPIO/

File: MCU_GPIO.kicad_sch

Title: Robot Wall-A

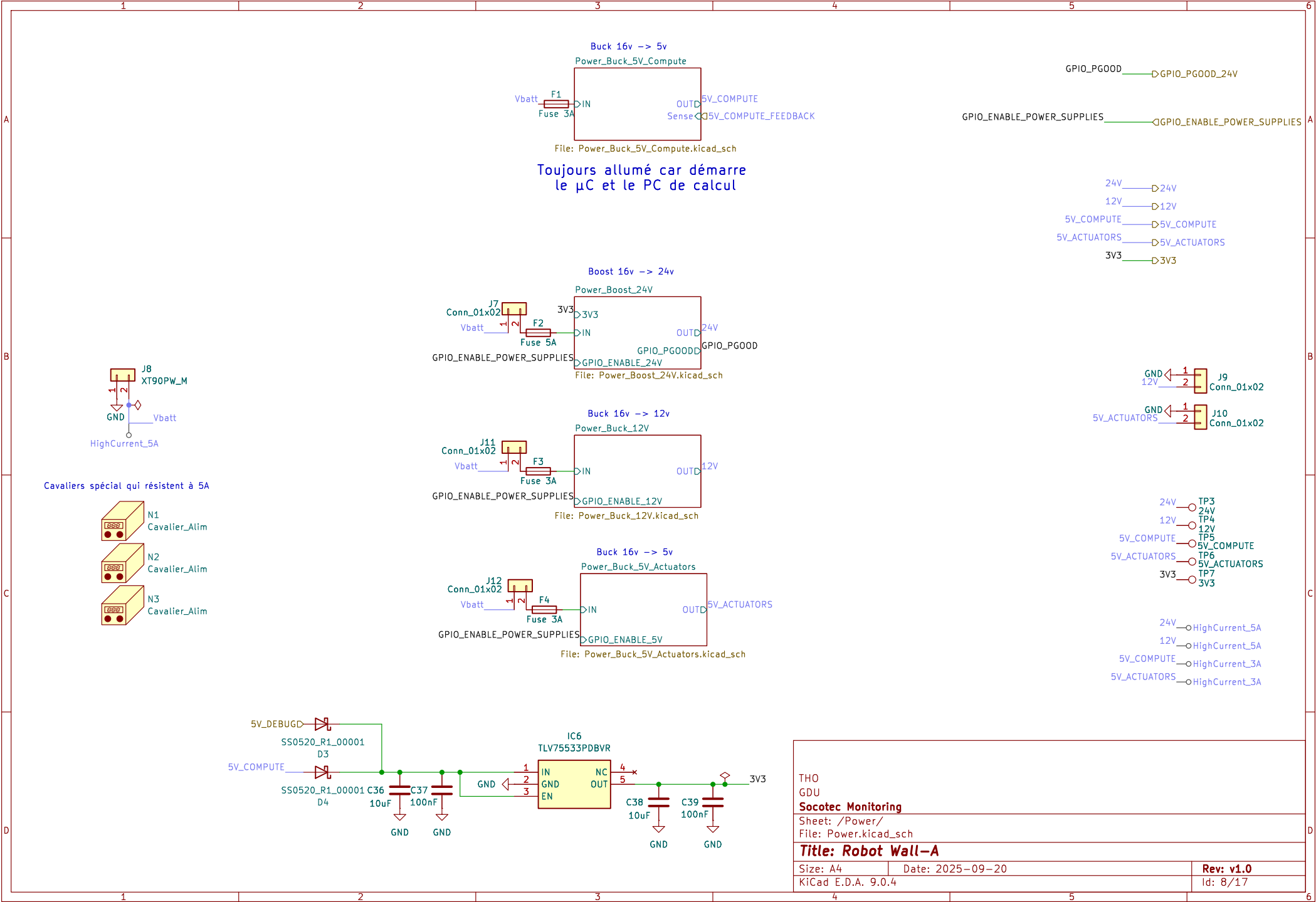
Size: A4

Date: 2025-09-20

Rev: v1.0

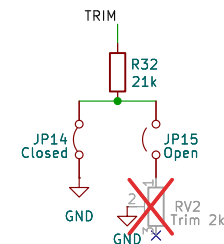
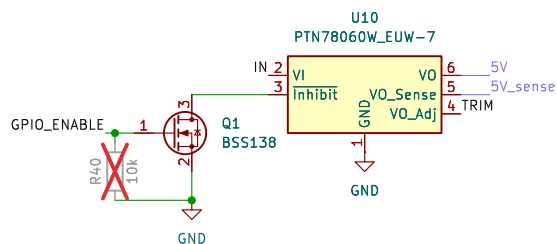
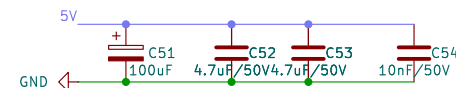
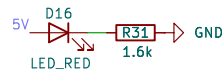
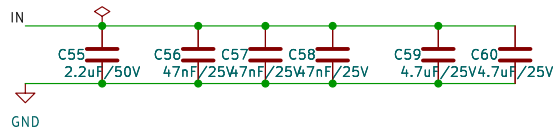
KiCad E.D.A. 9.0.4

Id: 7/17



IND IN

5V OUT
5V_sense Sense



TH0
GDU

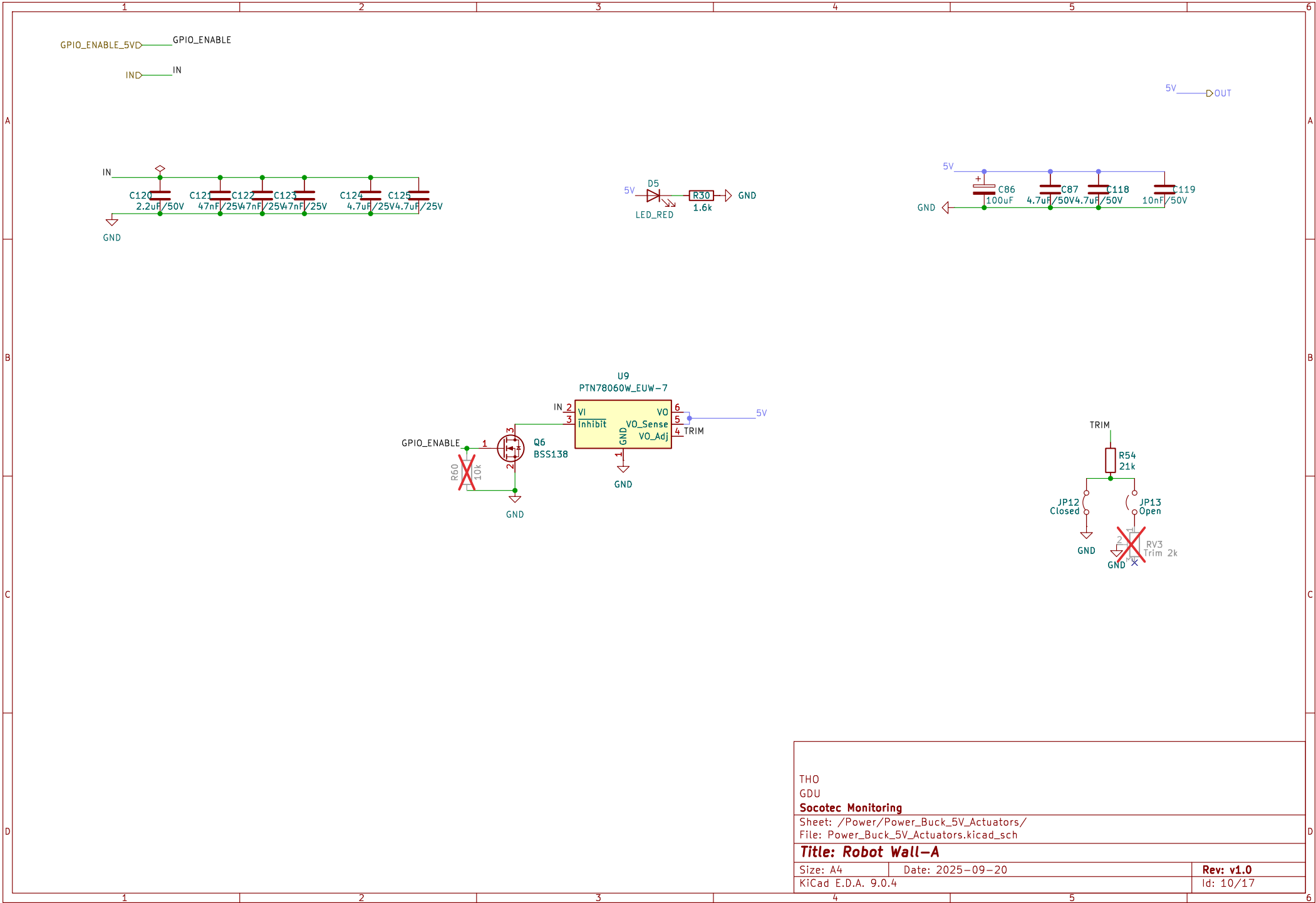
Socotec Monitoring

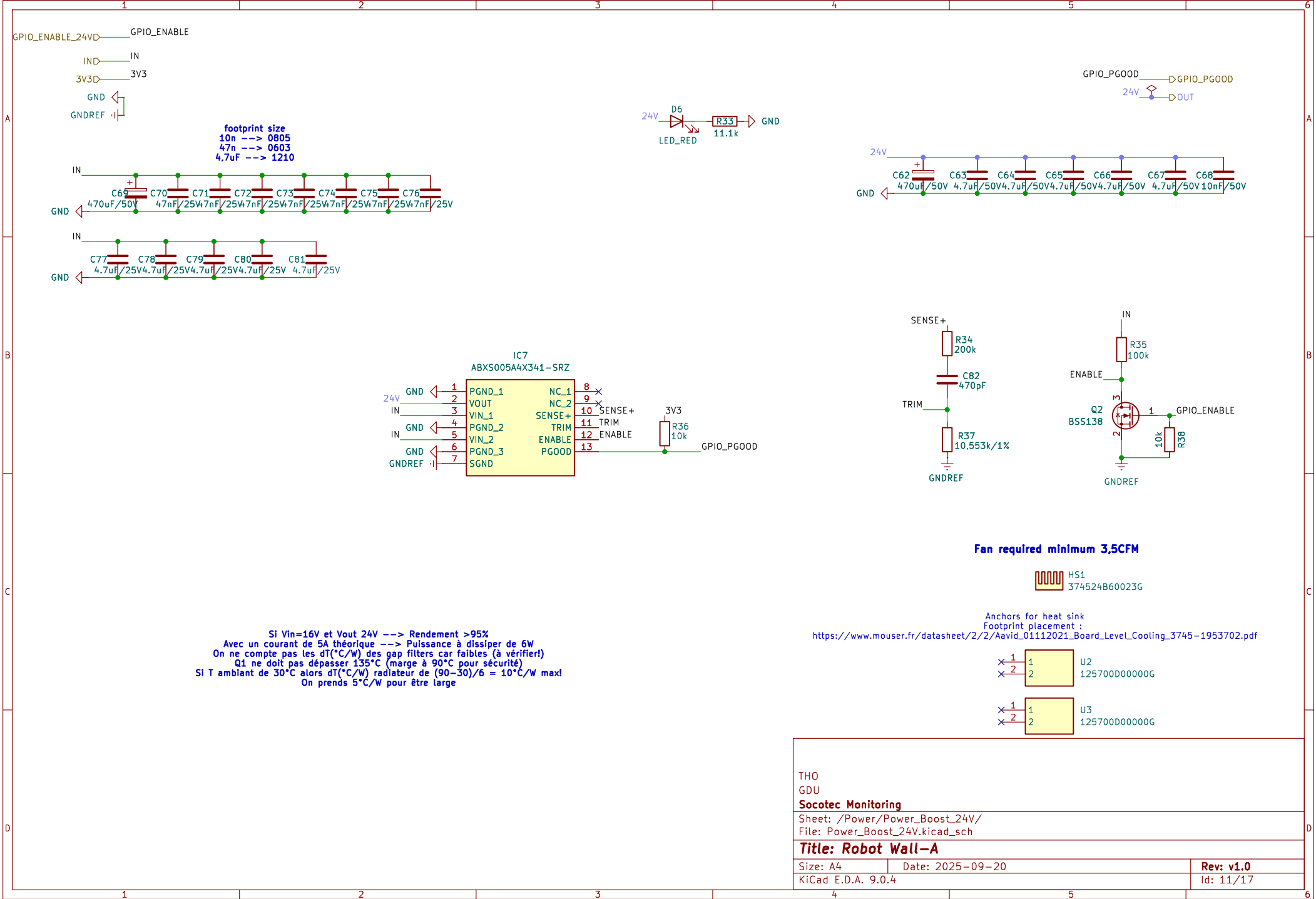
Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

Title: Robot Wall-A

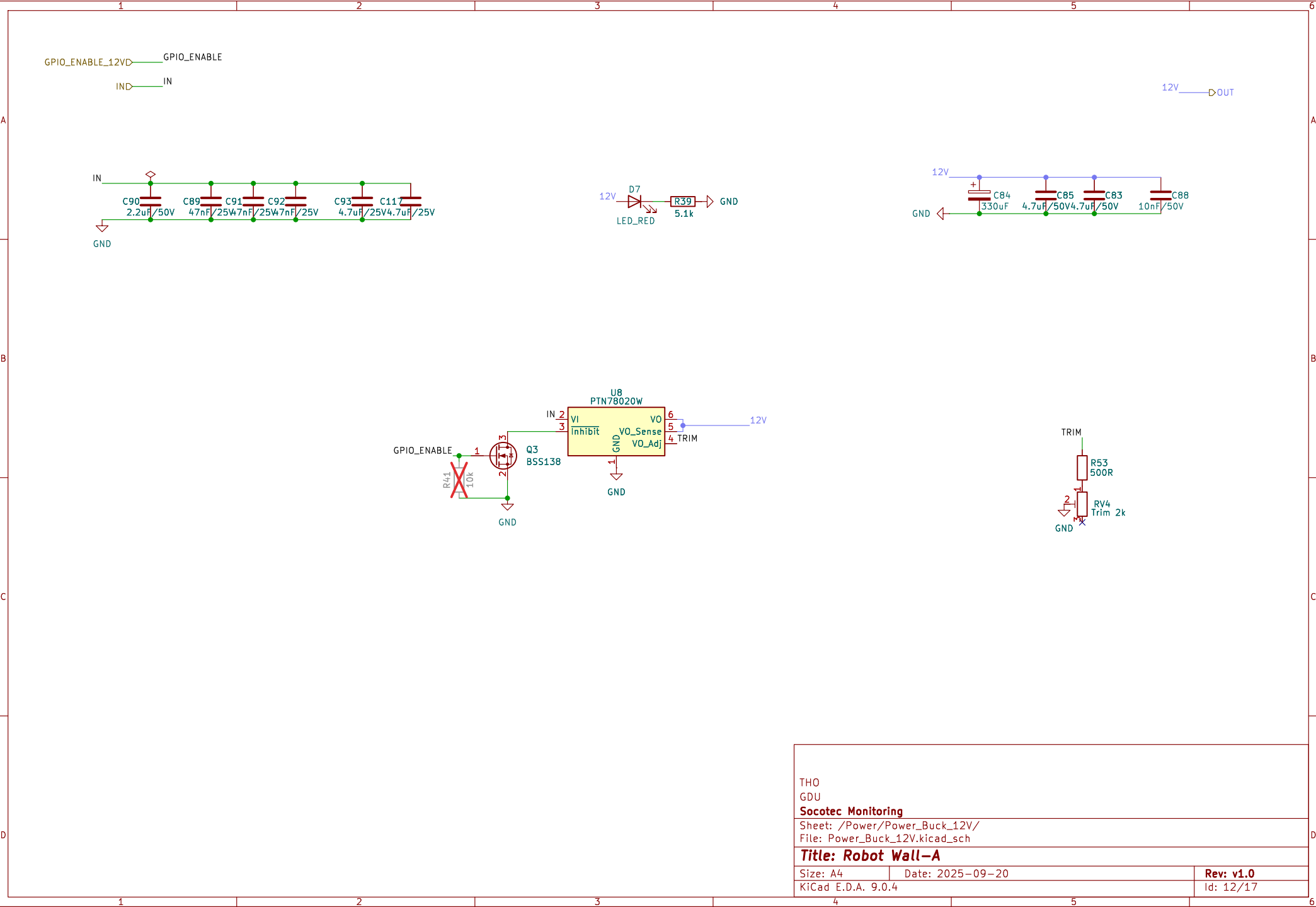
Size: A4 Date: 2025-09-20
KiCad E.D.A. 9.0.4

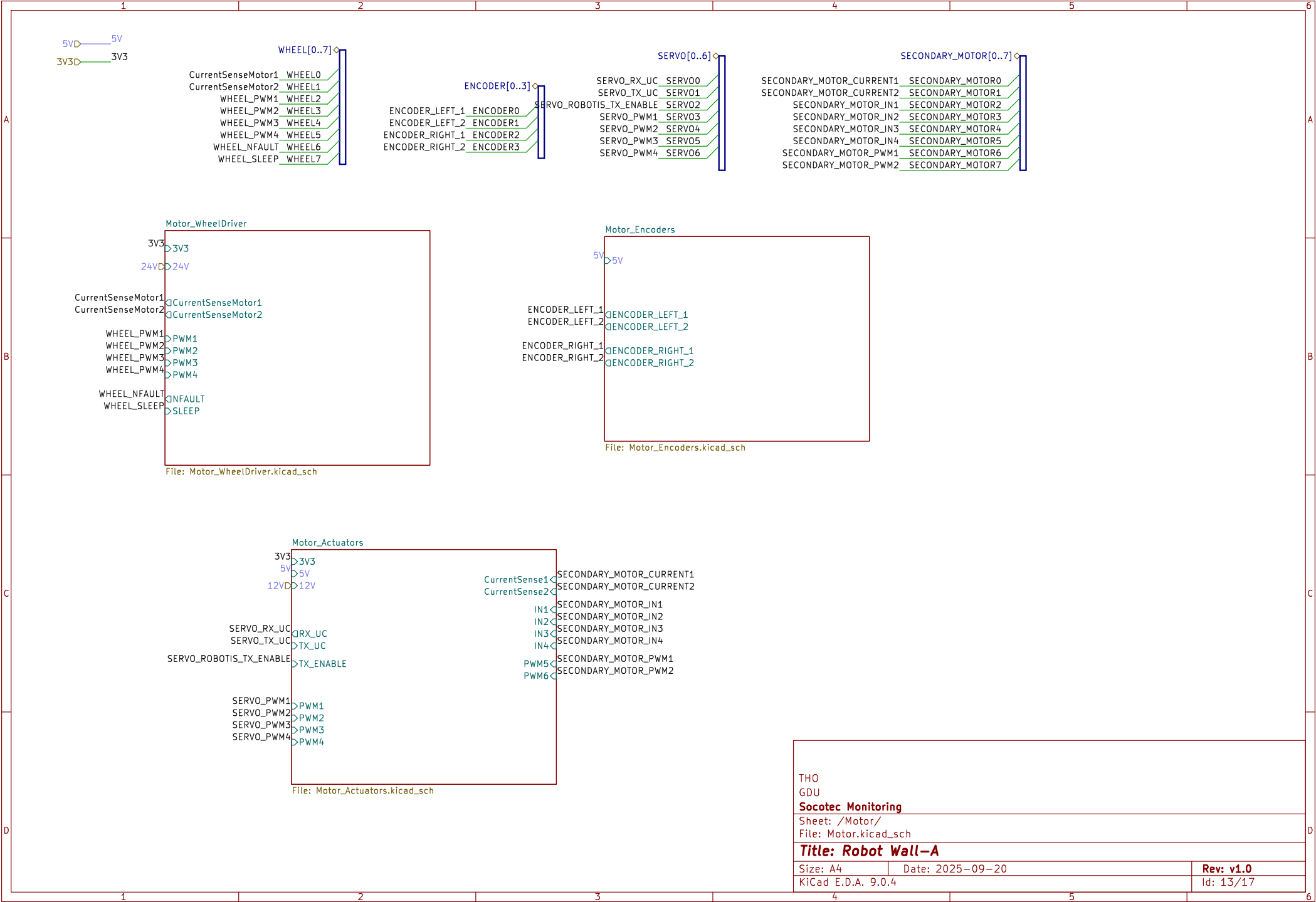
Rev: v1.0
Id: 9/17





Si Vin=16V et Vout 24V --> Rendement >95%
Avec un courant de 5A théorique --> Puissance à dissiper de 6W
On ne compte pas les dT(*C/W) des gap filters car faibles (à vérifier!)
Q1 ne doit pas dépasser 135°C (marge à 90°C pour sécurité)
Si T ambiant de 30°C alors dT(*C/W) radiateur de (90-30)/6 = 10°C/W max!
On prends 5°C/W pour être large



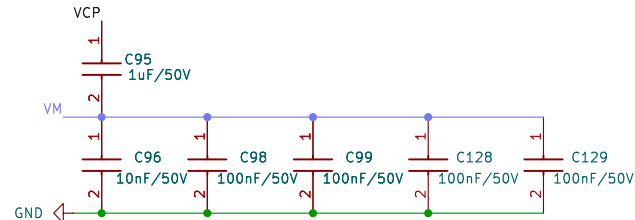
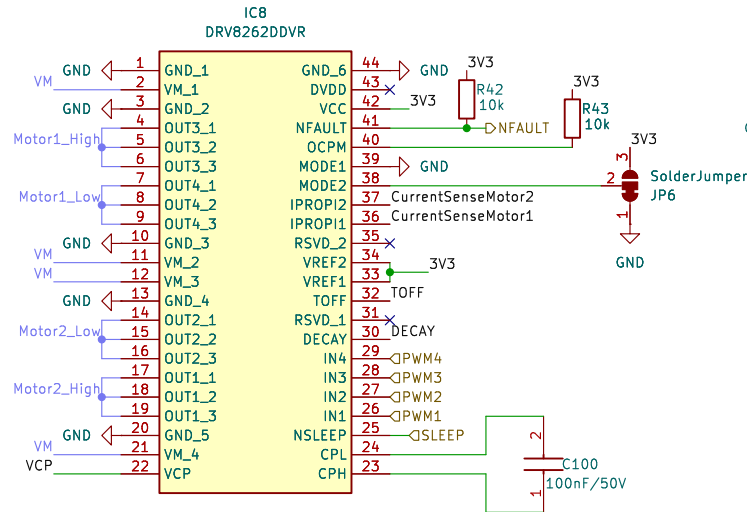


3V3D 3V3
24VD VM

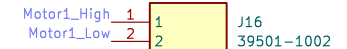
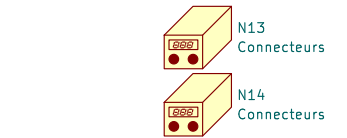
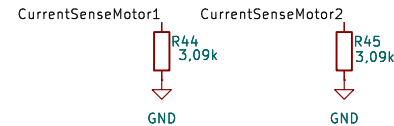


Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

CurrentSenseMotor1 > CurrentSenseMotor1
CurrentSenseMotor2 > CurrentSenseMotor2

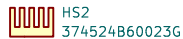
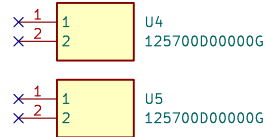


Current limit to 5A (example page 21 datasheet)



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

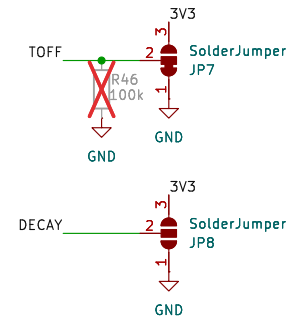


Fan required minimum 3,5CFM

2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High > HighCurrent_5A
Motor1_Low > HighCurrent_5A
Motor2_High > HighCurrent_5A
Motor2_Low > HighCurrent_5A

THO
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

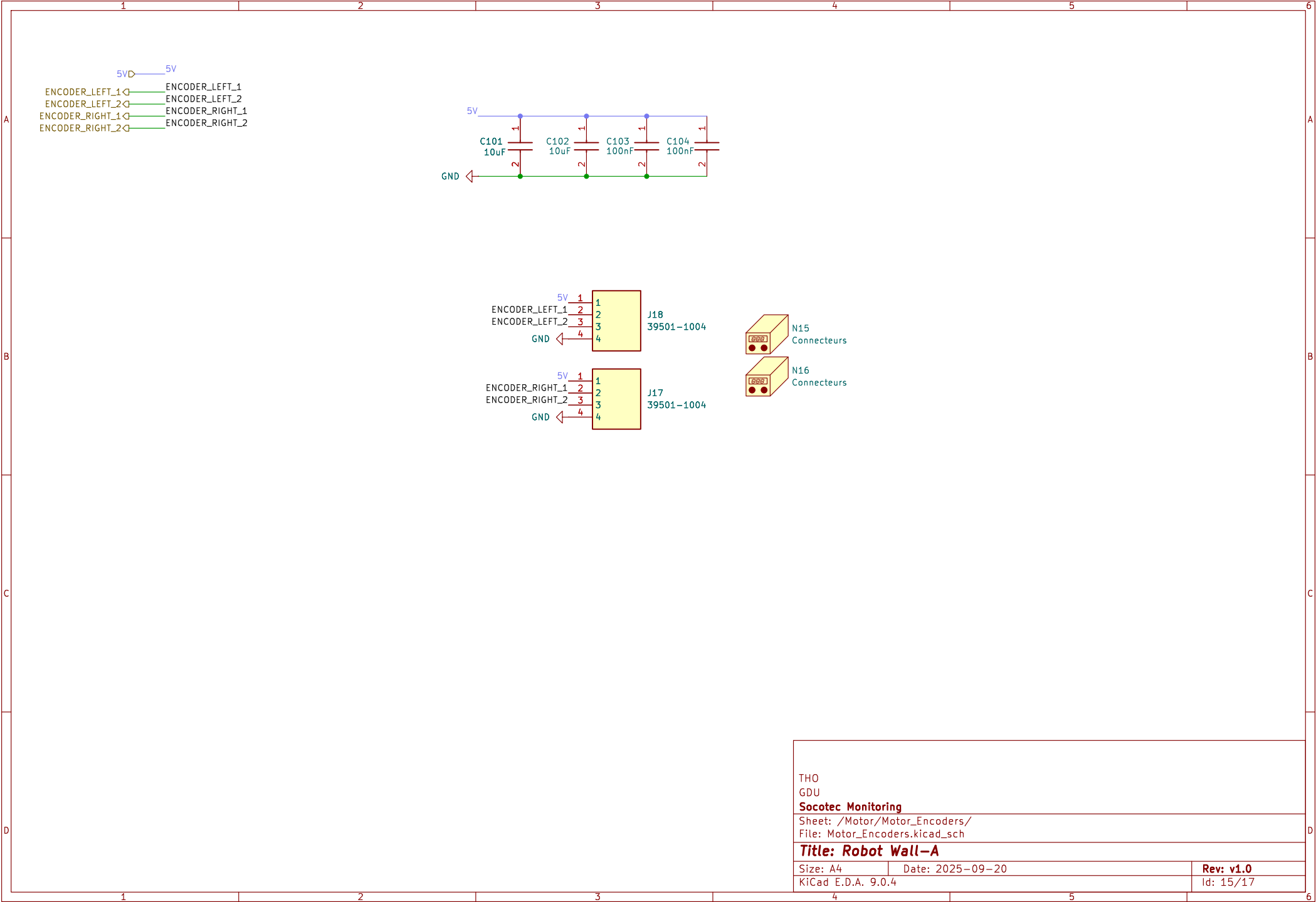
Title: Robot Wall-A

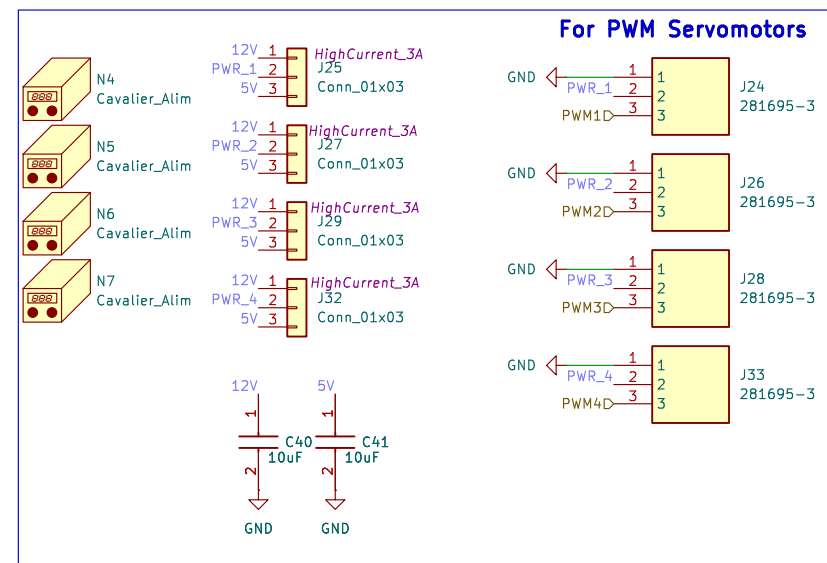
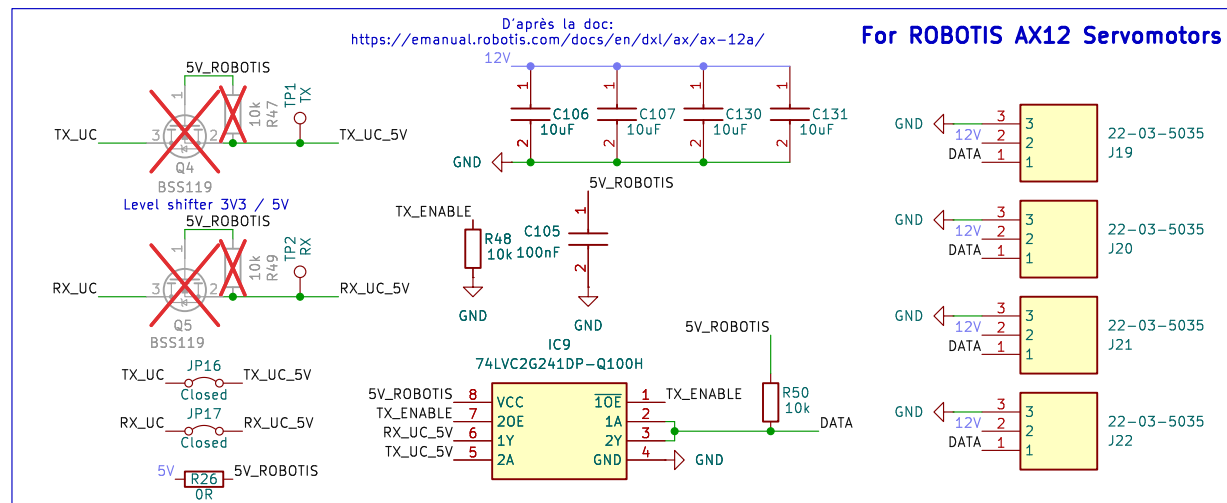
Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

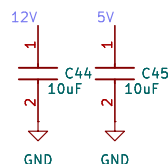
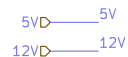
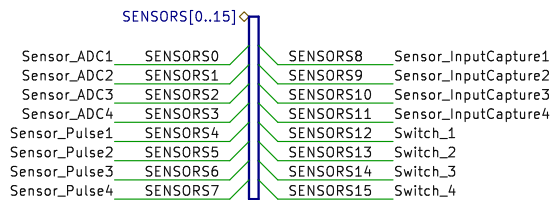
Rev: v1.0

Id: 14/17



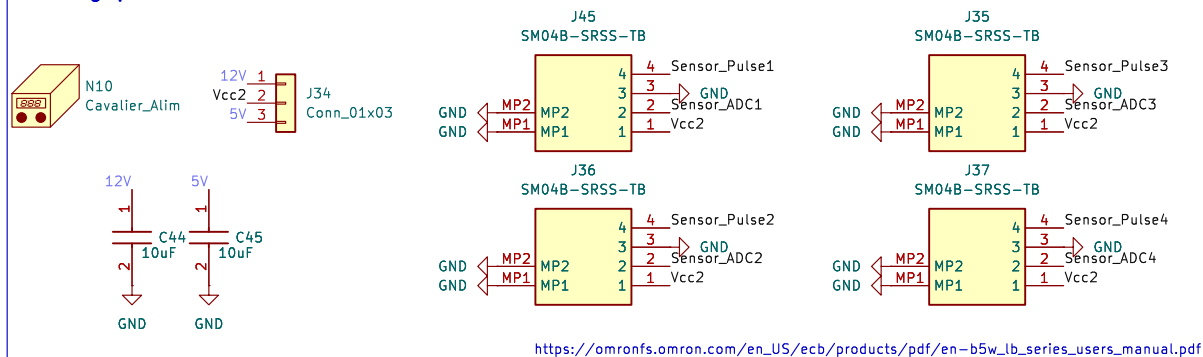


KiCad E.D.A. 9.0.4

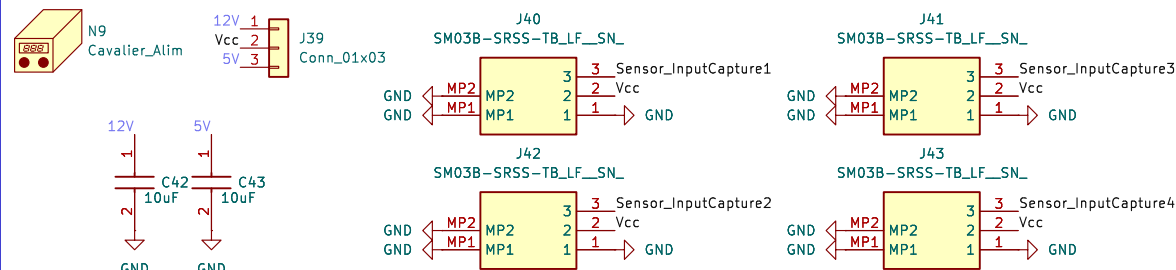


Capteurs de distance

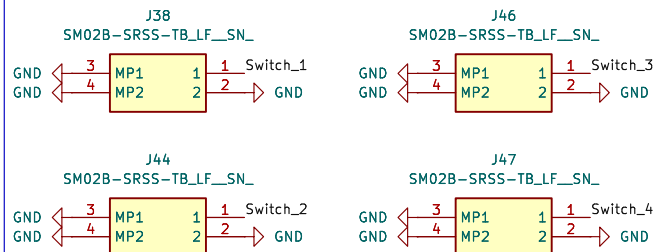
Pour capteur B5W-LB2101-1 analogique 5V



Pour capteur Pololu <https://www.pololu.com/product/5464>



Interrupteurs standard



TH0
GDU

Socotec Monitoring

Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20
KiCad E.D.A. 9.0.4

Rev: v1.0
Id: 17/17